1. Generate a correct time-based six-digit code with seconds remaining

For each saved authenticator account, the system calculates a six-digit code from the account's stored secret and the current time, using the standard 30-second time-based method. The code must match what other authenticator apps produce for the same secret. The screen shows the code along with the seconds remaining before it changes, and it refreshes automatically when a new code starts.

2. Save files atomically with owner-only access so a crash never corrupts data

Every save writes to a temporary file first, flushes it to disk, then renames it over the real file. If the program or computer stops partway through, the old complete file or the new complete file remains, never a half-written one. The data folder and every file in it are readable only by the owner (permissions 0700 and 0600), and the program refuses to use a folder or file that other users can access.

3. Wipe keys and plaintext from memory after use

Keys, the master password, and decrypted data are held in buffers that overwrite themselves with zeros as soon as they are no longer needed. This also happens when the vault is locked or the program closes. The wipe uses a method the compiler cannot skip, and secrets are kept in these buffers instead of ordinary strings so no stray copies are left behind.

4. Hide typed passwords and keep secrets out of logs and error messages

Passwords are shown as dots while the user types them, with an optional button to reveal them temporarily. Logs, error messages, and crash output never contain passwords, keys, or saved entries. Errors stay generic, such as "incorrect password" or "file damaged," so nothing about the secret data is revealed.

5. Add a password entry

The user can save a new login by entering a site name, a username, and a password. The program checks that the site name is not empty and rejects entries that are too long or contain control characters. The entry is saved to the encrypted passwords file before the screen reports success. If the save fails, the change is undone and the user sees a clear message.

6. View and list password entries

The user can see a list of all saved entries showing the site and username, with the passwords hidden by default. Selecting an entry shows its details, and the password is revealed only when the user asks for it. If the passwords file fails its integrity check, the list shows an error and nothing is loaded.

7. Generate a random password with a chosen length and character types

The user picks a length and which character types to include (uppercase, lowercase, digits, symbols), and the program creates a password using a secure random source, not a predictable one. It guarantees at least one character from each chosen type and rejects settings that cannot produce a valid password, such as no types selected or a length too short. The user can then copy the result or save it straight into a new entry.